Communicating with Models and Data

Hassan Sayed, Market Shaping Accelerator
(Center for Global Development × Dartmouth × UChicago)
Midwest Theory Conference,
hsayed@cgdev.org

October 17, 2026

A policymaker needs help designing a carbon tax. What tax should she choose?

- Nonlinear relationship: low taxes don't curb emissions, high taxes generate costs.
- She hires a **consultant** to help her determine what tax maximizes welfare.

A policymaker needs help designing a carbon tax. What tax should she choose?

- Nonlinear relationship: low taxes don't curb emissions, high taxes generate costs.
- She hires a **consultant** to help her determine what tax maximizes welfare.

Consultant collects data, estimates **model** mapping **tax** to welfare. Potentially valuable...

A policymaker needs help designing a carbon tax. What tax should she choose?

- Nonlinear relationship: low taxes don't curb emissions, high taxes generate costs.
- She hires a **consultant** to help her determine what tax maximizes welfare.

Consultant collects data, estimates **model** mapping **tax** to welfare. Potentially valuable...

...but three issues:

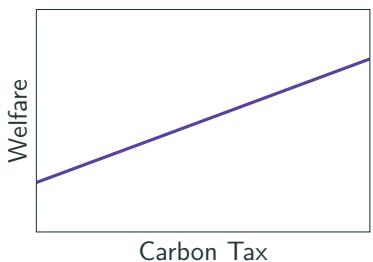
A policymaker needs help designing a carbon tax. What tax should she choose?

- Nonlinear relationship: low taxes don't curb emissions, high taxes generate costs.
- She hires a **consultant** to help her determine what tax maximizes welfare.

Consultant collects data, estimates **model** mapping **tax** to welfare. Potentially valuable...

...but three issues:

1. Model **too simple**. E.g. **linear regression** of welfare on taxes **neglects nonlinear tradeoff**.



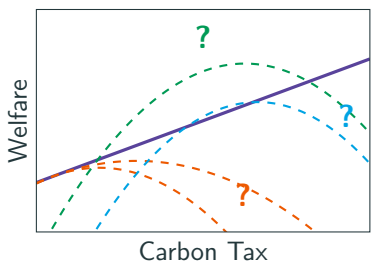
A policymaker needs help designing a carbon tax. What tax should she choose?

- Nonlinear relationship: low taxes don't curb emissions, high taxes generate costs.
- She hires a **consultant** to help her determine what tax maximizes welfare.

Consultant collects data, estimates **model** mapping **tax** to welfare. Potentially valuable...

...but three issues:

1. Model **too simple**. E.g. **linear regression** of welfare on taxes **neglects nonlinear tradeoff**.



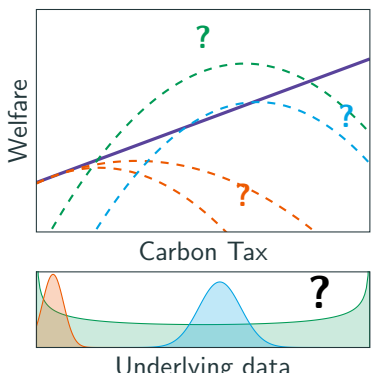
A policymaker needs help designing a carbon tax. What tax should she choose?

- Nonlinear relationship: low taxes don't curb emissions, high taxes generate costs.
- She hires a **consultant** to help her determine what tax maximizes welfare.

Consultant collects data, estimates **model** mapping **tax** to welfare. Potentially valuable...

...but three issues:

1. Model **too simple**. E.g. **linear regression** of welfare on taxes **neglects nonlinear tradeoff**.
2. **Underlying data opaque** or proprietary; only informative about certain choices.



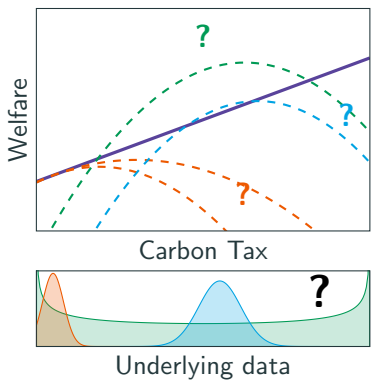
A policymaker needs help designing a carbon tax. What tax should she choose?

- Nonlinear relationship: low taxes don't curb emissions, high taxes generate costs.
- She hires a **consultant** to help her determine what tax maximizes welfare.

Consultant collects data, estimates **model** mapping **tax** to welfare. Potentially valuable...

...but three issues:

1. Model **too simple**. E.g. **linear regression** of welfare on taxes **neglects nonlinear tradeoff**.
2. **Underlying data opaque** or proprietary; only informative about certain choices.
3. Consultant wants her to be **optimistic** about how much she can improve welfare (vs. what happens).



What this Paper Studies

Many decisions are informed by **models** or **narratives** collected by others:

- **Organizations**: consultants use expertise to provide **scenarios** to CEOs;
- **Finance**: fund managers present models of investment returns to clients;
- **Politics**: politicians provide **narratives** about effects of new laws;
- **AI**: LLM chatbots uses internal data to provide recommendations.

How do **strategic model simplification and data collection** shape what we learn from externally provided models?

Receiver must make choice, hires sender to provide **model** mapping choices to outcomes.

- Sender collects data, estimates **accurate** or **simple** model.
- **Simple** model **fixes curvature** of relationship between choices and outcomes.
- Sender wants to make receiver optimistic: maximize **perceived** expected utility.

Receiver must make choice, hires sender to provide **model** mapping choices to outcomes.

- Sender collects data, estimates **accurate** or **simple** model.
- **Simple** model **fixes curvature** of relationship between choices and outcomes.
- Sender wants to make receiver optimistic: maximize **perceived** expected utility.

I show that **model simplicity** creates incentives to collect data irrelevant to receiver's choice.

- Underlying data contain info. about **extreme choices** that will never be made.
- When decisions are multivariate, underlying data is highly **collinear**.

Receiver must make choice, hires sender to provide **model** mapping choices to outcomes.

- Sender collects data, estimates **accurate** or **simple** model.
- **Simple** model **fixes curvature** of relationship between choices and outcomes.
- Sender wants to make receiver optimistic: maximize **perceived** expected utility.

I show that **model simplicity** creates incentives to collect data irrelevant to receiver's choice.

- Underlying data contain info. about **extreme choices** that will never be made.
- When decisions are multivariate, underlying data is highly **collinear**.

I then show that the **market for senders** is plagued by simple models and **polarization**.

- Accurate sender benefits if receiver thinks his model is simple $\implies$ surrounds himself with simple senders who estimate **excessively low or high curvature** models.
- Limiting cases: simple senders estimate **linear regressions** or **worst-case scenarios**.

Literature

Learning from models: Eliaz and Spiegler (2020), Schwartzstein and Sunderam (2021), Levy et al. (2022b), and Montiel Olea et al. (2022)

Strategic data collection: Henry and Ottaviani (2019), Di Tillio et al. (2021), and Dai et al. (2026).

Gains from simple models: Schwartzstein and Sunderam (2021), Aina (2023), Barron and Fries (2024), and Aina and Schneider (2025)

Delegation to experts: Li and Wang (2018), Deimen and Szalay (2019), Kishishita (2020), Cheng and Hsiaw (2022), and Kolotilin and Zapechelnyuk (2025)

Misspecified models: Alonso and Câmara (2016), Esponda and Pouzo (2016), Spiegler (2016), Heidhues et al. (2018), Mailath and Samuelson (2020), Bohren and Hauser (2021), Fudenberg et al. (2021), De Clippel and Zhang (2022), Gleyze and Pernoud (2022), Levy et al. (2022a), Levy et al. (2022b), Bohren and Hauser (2025), and Eliaz et al. (2025)

Receiver

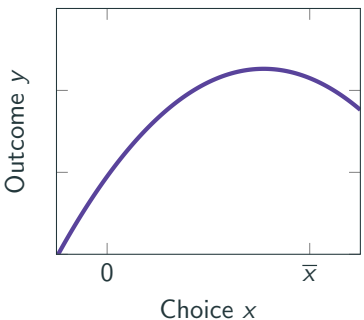
Receiver R makes choice $x \in [0, \overline{x}] = X$ to maximize outcome y. R is Bayesian.

R has prior over DGPs/"models" $y = f_R(x)$:

$$f_R(x) = \alpha_0 + \alpha_1 x - \gamma x^2, \qquad \alpha_0, \alpha_1 \in \mathbb{R},\ \gamma \in [0, \overline{\gamma}].$$

- Concave quadratic, **don't know** $\alpha_0, \alpha_1, \gamma$.
- γ: **curvature** of cost or tradeoff.
- Cannot interpret raw data on her own. But can hire an **expert** sender S to help.

Sender supplies **model**/**story** about effects of choices.



Sender

S collects data, estimates model. His type θ determines whether model **accurate** or **simple**.

- **Accurate sender**: type a. Estimates models F_a, same as R:

$$f_S^a(x) = \beta_0 + \beta_1 x - \delta x^2, \qquad \beta_0, \beta_1 \in \mathbb{R}, \qquad \delta \in [0, \overline{\gamma}].$$

  Prob. of seeing type a is p_a. (Bayesian over F_a).

Sender

S collects data, estimates model. His type θ determines whether model **accurate** or **simple**.

- **Accurate sender**: type a. Estimates models F_a, same as R:

$$f_S^a(x) = \beta_0 + \beta_1 x - \delta x^2, \qquad \beta_0, \beta_1 \in \mathbb{R}, \qquad \delta \in [0, \overline{\gamma}].$$

  Prob. of seeing type a is p_a. (Bayesian over F_a).

- **Simple sender**: type δ. Estimates models F_δ with **restricted curvature**:

$$f_S^\delta(x) = \beta_0 + \beta_1 x - \delta x^2, \qquad \beta_0, \beta_1 \in \mathbb{R}, \qquad \delta \text{ fixed}$$

  Prob. of seeing type δ is $(1 - p_a)t(\delta)$ for distribution t on $[0, \overline{\gamma}]$. (Bayesian over F_δ).

Sender

S collects data, estimates model. His type θ determines whether model **accurate** or **simple**.

- **Accurate sender**: type a. Estimates models F_a, same as R:

$$f_S^a(x) = \beta_0 + \beta_1 x - \delta x^2, \qquad \beta_0, \beta_1 \in \mathbb{R}, \qquad \delta \in [0, \overline{\gamma}].$$

  Prob. of seeing type a is p_a. (Bayesian over F_a).

- **Simple sender**: type δ. Estimates models F_δ with **restricted curvature**:

$$f_S^\delta(x) = \beta_0 + \beta_1 x - \delta x^2, \qquad \beta_0, \beta_1 \in \mathbb{R}, \qquad \delta \text{ fixed}$$

  Prob. of seeing type δ is $(1 - p_a)t(\delta)$ for distribution t on $[0, \overline{\gamma}]$. (Bayesian over F_δ).

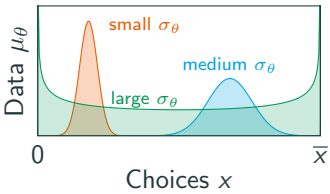
- $\delta = 0$: S models **linear regressions**. $\delta = \overline{\gamma}$: **worst-case scenario** model.

Simple sender is **calibrating an elasticity** to a fixed value, vs. organically estimating it.

Sender Data Sampling and Model Fitting

S samples data as follows. Chooses distribution μ_θ of x values, observes corresponding y values.

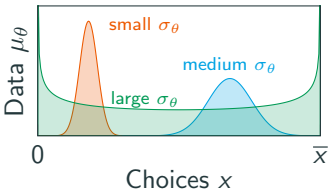
- μ_θ is symmetric
- Mean of μ_θ is $x_\theta \in X$.
- Variance of μ_θ is $\sigma_\theta^2 \in [\underline{\sigma}^2, \overline{\sigma}^2]$ for $\underline{\sigma}$ small, $\overline{\sigma}$ large.



Sender Data Sampling and Model Fitting

S samples data as follows. Chooses distribution μ_θ of x values, observes corresponding y values.

- μ_θ is symmetric
- Mean of μ_θ is $x_\theta \in X$.
- Variance of μ_θ is $\sigma_\theta^2 \in [\underline{\sigma}^2, \overline{\sigma}^2]$ for $\underline{\sigma}$ small, $\overline{\sigma}$ large.



Choice of $\mu_\theta \implies$ data $D_\mu^\theta = \{(y, x)\}$. Estimates **model of best fit**:

$$f_S^\theta = \arg\min_{f \in F_\theta} \mathbb{E}_{D_\mu^\theta}[(y - f(x))^2].$$

Type a provides true model, type δ misspecified model. μ_θ and D_μ^θ **unobserved by receiver**.

- R can't interpret raw data. This is the point of hiring S.
- S's data may be proprietary, or embed expertise.
- S can't just make things up (third-party audit, "replication package").

R's and S's Utility

R maximizes expected value of y. Given S's model f_S, R updates over f_R, chooses

$$x^*(f_S) = \arg\max_x \mathbb{E}_{f_R|f_S}[f_R(x)]$$

Given a model f_S, S's utility from R's choice is:

$$U_S(f_S) = \mathbb{E}_{f_R|f_S}[f_R(x^*(f_S))].$$

S maximizes R's **perceived expected utility**: wants to make her optimistic.

- Consultant "sells story" to policymaker/CEO so that she adopts a program.
- Politician "sells narrative" to voters so that they vote for her.

S doesn't care what actually happens, only about R's perception of what might happen.

Timing

1. Nature draws S's type θ and the true DGP f_R.
2. S chooses μ^*_θ, collects data, calculates model of best fit f_S.
3. S provides f_S to R, R updates over true set of DGPs, S's type, S's data collection.
4. R makes her decision $x^*(f_S)$.
5. S receives utility $\mathbb{E}_{f_R|f_S}[f_R(x^*(f_S))]$.
6. R receives utility $f_R(x^*(f_S))$.

Learning: R entertains S's model, but also others

Proposition 1.

Suppose S provides $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$. Set of models R considers are given by

$$\left\{ f_R(x) = f_S(x) + (\gamma - \delta)(\sigma_\delta^2 - (x - x_\delta^2)) \quad : \quad \gamma \in [0, \overline{\gamma}] \right\}$$

where x_δ and σ_δ^2 are mean, variance of μ_δ.

All models have same mean, average slope across data. All models have same slope at x_δ.

Learning: R entertains S's model, but also others

Proposition 1.

Suppose S provides $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$. Set of models R considers are given by

$$\left\{ f_R(x) = f_S(x) + (\gamma - \delta)(\sigma_\delta^2 - (x - x_\delta^2)) \quad : \quad \gamma \in [0, \overline{\gamma}] \right\}$$

where x_δ and σ_δ^2 are mean, variance of μ_δ.

All models have same mean, average slope across data. All models have same slope at x_δ.

Linear models: $\beta_0 + \beta_1 x$

Learning: R entertains S's model, but also others

Proposition 1.

Suppose S provides $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$. Set of models R considers are given by

$$\left\{ f_R(x) = f_S(x) + (\gamma - \delta)(\sigma_\delta^2 - (x - x_\delta^2)) \quad : \quad \gamma \in [0, \overline{\gamma}] \right\}$$

where x_δ and σ_δ^2 are mean, variance of μ_δ.

All models have same mean, average slope across data. All models have same slope at x_δ.

Linear models: $\beta_0 + \beta_1 x + \gamma(\sigma_\delta^2 - (x - x_\delta^2))$, where $\gamma > 0$.

Learning: R entertains S's model, but also others

Proposition 1.

Suppose S provides $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$. Set of models R considers are given by

$$\left\{ f_R(x) = f_S(x) + (\gamma - \delta)(\sigma_\delta^2 - (x - x_\delta^2)) \quad : \quad \gamma \in [0, \overline{\gamma}] \right\}$$

where x_δ and σ_δ^2 are mean, variance of μ_δ.

All models have same mean, average slope across data. All models have same slope at x_δ.

Linear models: $\beta_0 + \beta_1 x + \gamma(\sigma_\delta^2 - (x - x_\delta^2))$, where $\gamma > 0$.

Worst-case models: $\beta_0 + \beta_1 x - \overline{\gamma} x^2$

Learning: R entertains S's model, but also others

Proposition 1.

Suppose S provides $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$. Set of models R considers are given by

$$\left\{ f_R(x) = f_S(x) + (\gamma - \delta)(\sigma_\delta^2 - (x - x_\delta^2)) \quad : \quad \gamma \in [0, \overline{\gamma}] \right\}$$

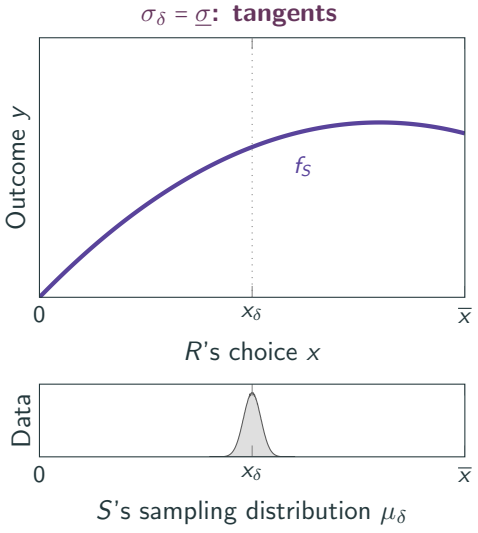
where x_δ and σ_δ^2 are mean, variance of μ_δ.

All models have same mean, average slope across data. All models have same slope at x_δ.

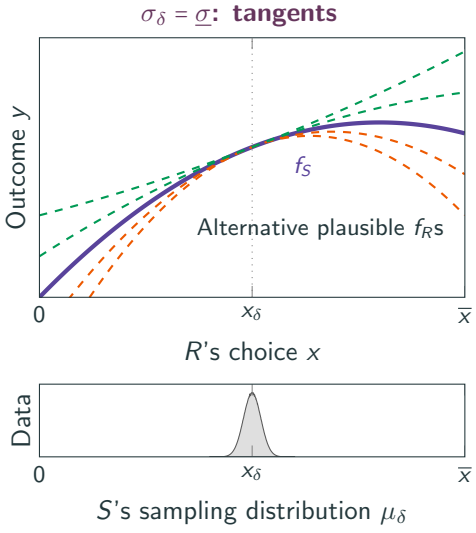
Linear models: $\beta_0 + \beta_1 x + \gamma(\sigma_\delta^2 - (x - x_\delta^2))$, where $\gamma > 0$.

Worst-case models: $\beta_0 + \beta_1 x - \overline{\gamma} x^2 + (\gamma - \overline{\gamma})(\sigma_\delta^2 - (x - x_\delta^2))$, where $(\gamma - \overline{\gamma}) < 0$.

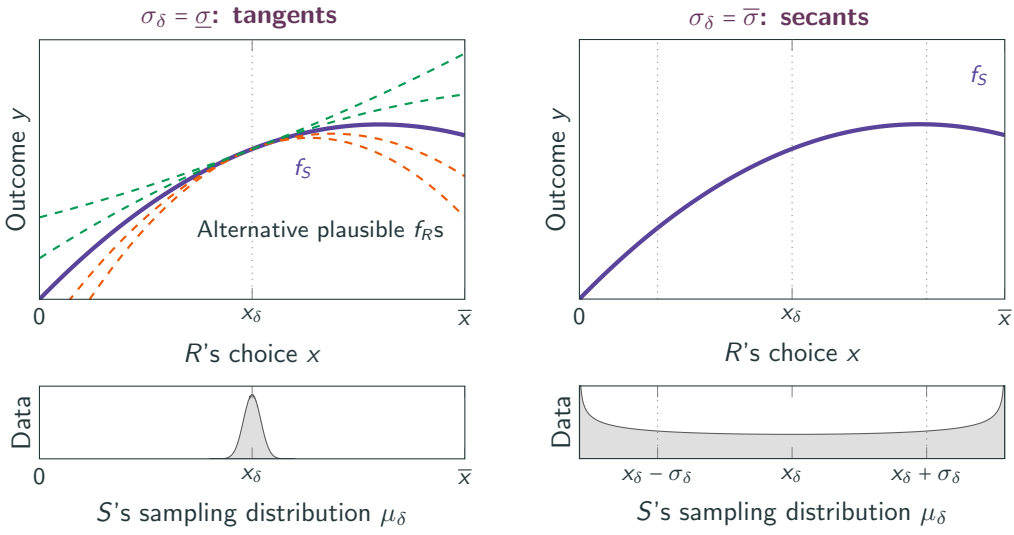
Plausible models: tangents vs. secants



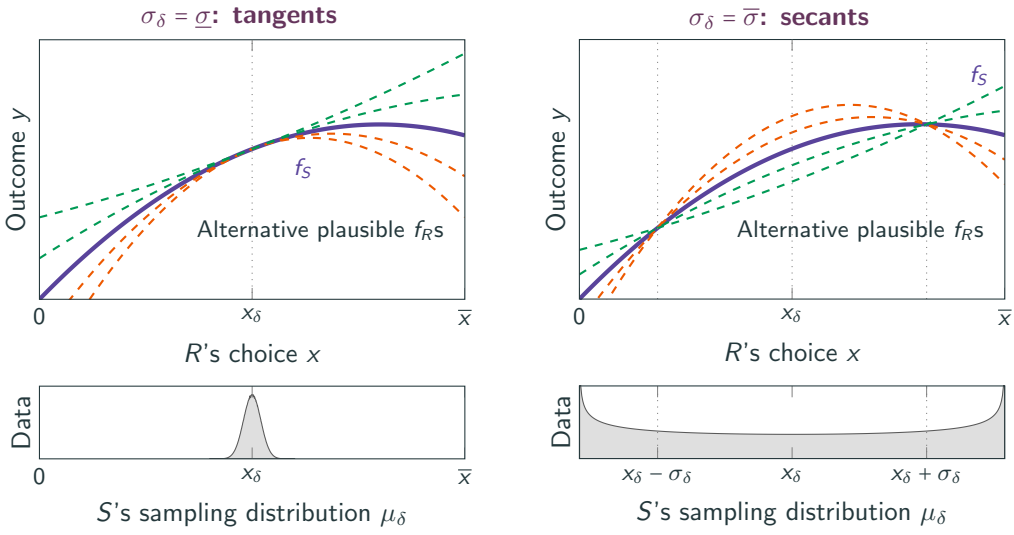
Plausible models: tangents vs. secants



Plausible models: tangents vs. secants



Plausible models: tangents vs. secants



Definitions and Result on Strategic Sampling Result

Two varieties of simple models: **high** and **low** curvature.

- **Low Curvature**: calibrates $\delta < \mathbb{E}_R^{\gamma}[\gamma]$ e.g. linear model.
- **High Curvature**: calibrates $\delta > \mathbb{E}_R^{\gamma}[\gamma]$ e.g. worst-case scenario model.

Definitions and Result on Strategic Sampling Result

Two varieties of simple models: **high** and **low** curvature.

- **Low Curvature**: calibrates $\delta < \mathbb{E}_R^{\gamma}[\gamma]$ e.g. linear model.
- **High Curvature**: calibrates $\delta > \mathbb{E}_R^{\gamma}[\gamma]$ e.g. worst-case scenario model.

Two sampling regimes.

- **Specialist**: $\sigma_\delta = \underline{\sigma} \approx 0$. "Local information."
- **Generalist**: $\sigma_\delta = \overline{\sigma}$. Excessively high variance distribution (e.g. bimodal).

Definitions and Result on Strategic Sampling Result

Two varieties of simple models: **high** and **low** curvature.

- **Low Curvature**: calibrates $\delta < \mathbb{E}_R^{\gamma}[\gamma]$ e.g. linear model.
- **High Curvature**: calibrates $\delta > \mathbb{E}_R^{\gamma}[\gamma]$ e.g. worst-case scenario model.

Two sampling regimes.

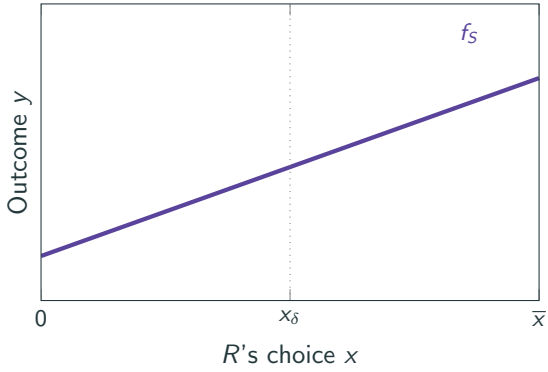
- **Specialist**: $\sigma_\delta = \underline{\sigma} \approx 0$. "Local information."
- **Generalist**: $\sigma_\delta = \overline{\sigma}$. Excessively high variance distribution (e.g. bimodal).

Theorem 1.

- **Low curvature sender**: *samples as* **generalist** $(\sigma_\delta = \overline{\sigma})$ *in center of* X $(x_\delta^* = \mathbb{E}_{f_S}^{\delta}[x^*(f_S)])$.
- **High curvature sender**: *samples as* **specialist** $(\sigma_\delta = \underline{\sigma})$ *at edge of* X $(x_\delta^* = 0 \text{ or } \overline{x})$.

Intuition: Low curvature Model, Specialist vs. Generalist Sampling

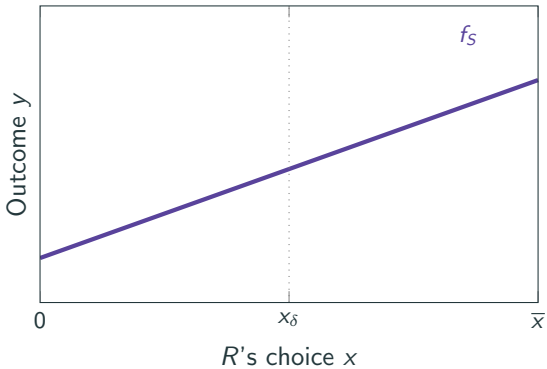
See f_S with $\delta < \mathbb{E}[\gamma]$.



Intuition: Low curvature Model, Specialist vs. Generalist Sampling

See f_S with $\delta < \mathbb{E}[\gamma]$.

- Could be true model or **low curvature** (e.g. linear).
- R entertains f_S and models that have steeper curvature.



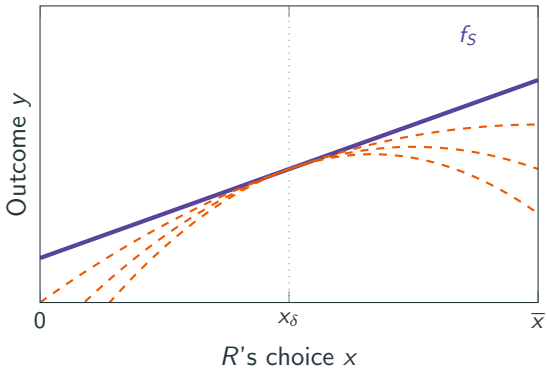
Intuition: Low curvature Model, Specialist vs. Generalist Sampling

See f_S with $\delta < \mathbb{E}[\gamma]$.

- Could be true model or **low curvature** (e.g. linear).
- R entertains f_S and models that have steeper curvature.

Specialist: other f_Rs **tangent** at x_δ.

- Steeper models lie **below** f_S.



Intuition: Low curvature Model, Specialist vs. Generalist Sampling

See f_S with $\delta < \mathbb{E}[\gamma]$.

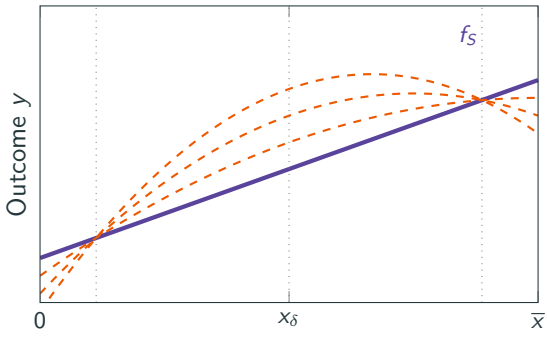
- Could be true model or **low curvature** (e.g. linear).
- R entertains f_S and models that have steeper curvature.

Specialist: other f_Rs **tangent** at x_δ.

- Steeper models lie **below** f_S.

Generalist: other f_Rs **secant**.

- **Vertically shifted up**
- R **more optimistic**.



Intuition: Low curvature Model, Specialist vs. Generalist Sampling

See f_S with $\delta < \mathbb{E}[\gamma]$.

- Could be true model or **low curvature** (e.g. linear).
- R entertains f_S and models that have steeper curvature.

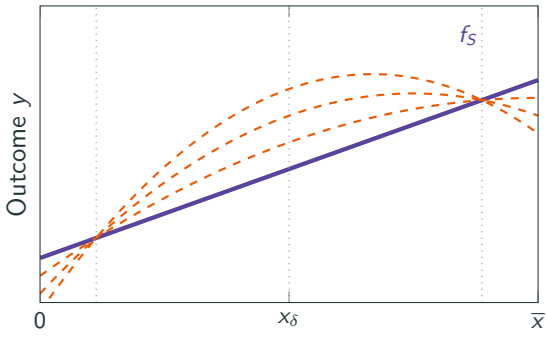
Specialist: other f_Rs **tangent** at x_δ.

- Steeper models lie **below** f_S.

Generalist: other f_Rs **secant**.

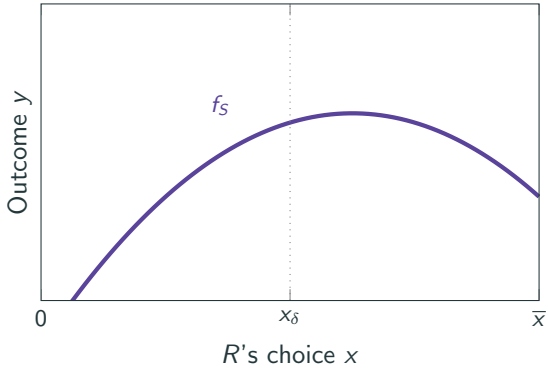
- **Vertically shifted up**
- R **more optimistic**.

Low curvature sender is **generalist**



Intuition: High Curvature Model, Specialist vs. Generalist Sampling

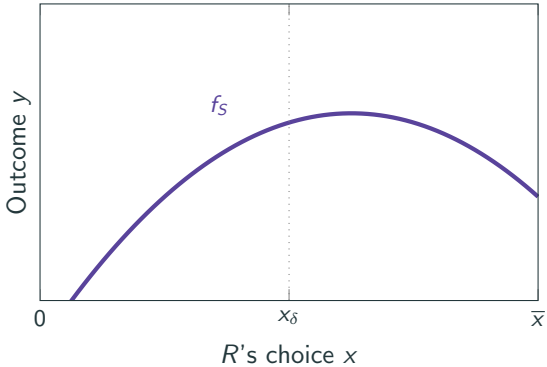
See f_S with $\delta > \mathbb{E}[\gamma]$.



Intuition: High Curvature Model, Specialist vs. Generalist Sampling

See f_S with $\delta > \mathbb{E}[\gamma]$.

- Could be true model or **high curvature** (e.g. worst-case).
- R entertains f_S and models that have flatter curvature.



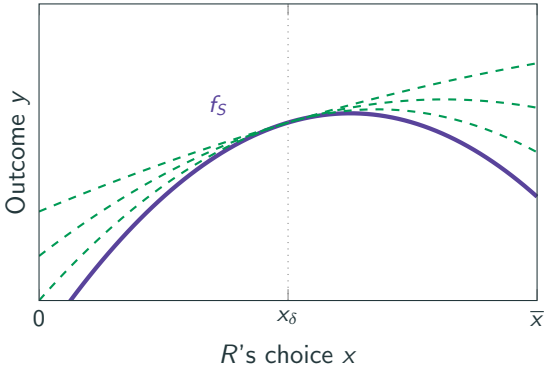
Intuition: High Curvature Model, Specialist vs. Generalist Sampling

See f_S with $\delta > \mathbb{E}[\gamma]$.

- Could be true model or **high curvature** (e.g. worst-case).
- R entertains f_S and models that have flatter curvature.

Specialist: other f_Rs **tangent** at x_δ.

- Flatter models lie **above** f_S.



Intuition: High Curvature Model, Specialist vs. Generalist Sampling

See f_S with $\delta > \mathbb{E}[\gamma]$.

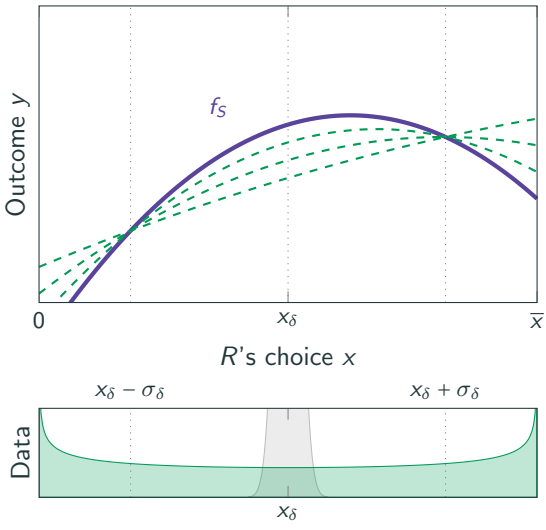
- Could be true model or **high curvature** (e.g. worst-case).
- R entertains f_S and models that have flatter curvature.

Specialist: other f_Rs **tangent** at x_δ.

- Flatter models lie **above** f_S.

Generalist: other f_Rs **secant**.

- **Vertically shifted down**.
- R **less optimistic**.



Intuition: High Curvature Model, Specialist vs. Generalist Sampling

See f_S with $\delta > \mathbb{E}[\gamma]$.

- Could be true model or **high curvature** (e.g. worst-case).
- R entertains f_S and models that have flatter curvature.

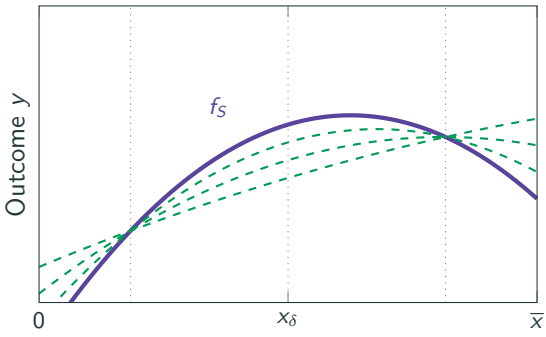
Specialist: other f_Rs **tangent** at x_δ.

- Flatter models lie **above** f_S.

Generalist: other f_Rs **secant**.

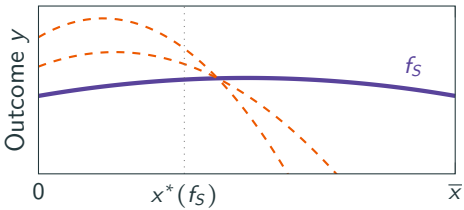
- **Vertically shifted down**.
- R **less optimistic**.

High curvature sender is **specialist**



Intuition: Where to Sample

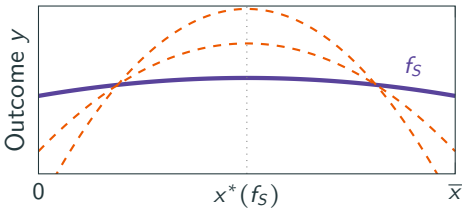
Low curvature model: generalist



- **Edge of X**: when $f_S'(x) = 0$, alternative models lie below.

Intuition: Where to Sample

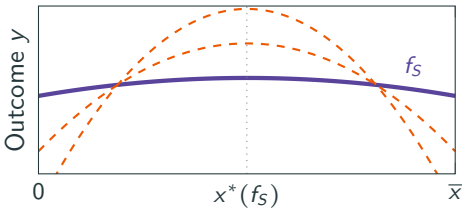
Low curvature model: generalist



- **Edge of X**: when $f_S'(x) = 0$, alternative models lie below.
- **Center of X**: when $f_S'(x) = 0$, alternative models lie above, large gain.

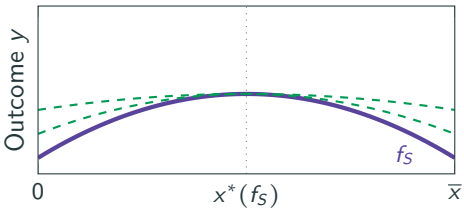
Intuition: Where to Sample

Low curvature model: generalist



- **Edge of X**: when $f_S'(x) = 0$, alternative models lie below.
- **Center of X**: when $f_S'(x) = 0$, alternative models lie above, large gain.

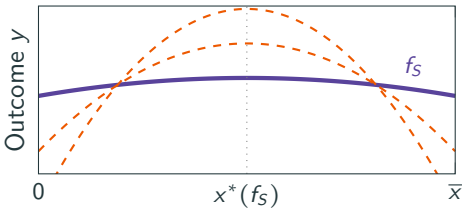
High curvature model: specialist



- **Center of X**: when $f_S'(x) = 0$, alternative models deliver same utility.

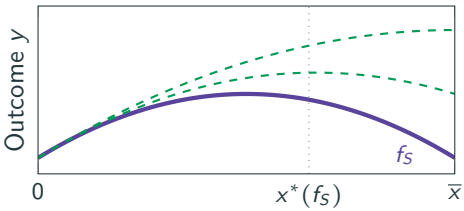
Intuition: Where to Sample

Low curvature model: generalist



- **Edge of** X: when $f_S'(x) = 0$, alternative models lie below.
- **Center of** X: when $f_S'(x) = 0$, alternative models lie above, large gain.

High curvature model: specialist



- **Center of** X: when $f_S'(x) = 0$, alternative models deliver same utility.
- **Edge of** X: when $f_S'(x) = 0$, alternative models lie far above f_S.

Irrelevant Data + Accurate Senders' Gain

Strategic data selection: data that are **likely irrelevant** to R's choice

- Low curvature: $x_\delta \pm \sigma_\delta$ may lie outside of $[0, \overline{x}]$. Data outside of R's choice set!
- High Curvature: precise info about "fringe choices" 0 or $\overline{x}$.

Given a model f_S, receiver's expected utility is

$$f_S(x) + \underbrace{\Pr(\theta = \delta | f_S)\left(\mathbb{E}^R_\gamma[\gamma|f_S] - \delta\right)\left(\sigma_\delta^2 - (x - x_\delta)^2\right)}_{\text{Gain due to simple senders}}$$

$\left(\mathbb{E}^R_\gamma[\gamma|f_S] - \delta\right)\left(\sigma_\delta^2 - (x - x_\delta)^2\right) \geq 0$ due to **strategic sampling**.

- If type a provides **true** f_S, R entertains models **more optimistic than truth**.
- Gain comes from being mistaken for **simple sender who strategically samples**.

Multivariate Choices

Imagine now $x \in \mathbb{R}^n$. Models given by $f_R : \mathbb{R}^N \to \mathbb{R}$:

$$f_R(x) = f_R(x) = \alpha_0 + \alpha' x - x' \gamma x \qquad \gamma \in \Gamma \text{ PSD}$$

Data sampled via μ_δ spherically symmetric: mean x_δ, covariance matrix σ_δ:

- Variances: $\sigma_i := \Sigma_{ii} \in [\underline{\sigma}_i, \overline{\sigma}_i]$.
- Correlations: $\rho_{ij} := \frac{\Sigma_{\delta,ij}}{\sigma_i \sigma_j} \in [-\overline{\rho}, \overline{\rho}]$

Type δ now assumes a PSD matrix δ. Can generalize the results from before.

- Type δ **neglects interactions** if $\mathbb{E}^R_\gamma[\gamma_{ij} + \gamma_{ji}] > \delta_{ij} + \delta_{ji}$.
- Otherwise, **exaggerates interactions**.

Multivariate Choices: Maximizing Collinearity

Theorem 2.

- *Senders who neglect interactions between x_i and x_j maximize correlation between them: $\rho_{ij}^* = \overline{\rho}$.*
- *Senders who exaggerate interactions between x_i and x_j maximize anticorrelation between them: $\rho_{ij}^* = -\overline{\rho}$.*

Key insight: **sample data** to **maximize collinearity** between variables.

- Example: use data where **carbon tax** and **subsidy** that are **in effect at same time**.
- **Heavily unbalanced data**. In finite samples, imprecise model parameter estimates.

The Market for Senders: Motivation

So far, have taken the distribution $t(\delta)$ of simple senders as exogenous.

- But a type a sender **gains** from being mistake for simple type.
- How would he **endogeneously choose** a distribution $t^*(\delta)$ of simple senders?

Many interpretations of this problem, depending on application.

- Senior consultant **hires** other consultants at organization who estimate simple models
- Fund manager deploys "robo-advisors," provide simple rules of thumb for investing
- Politician propagates **simple narratives** of effects of new laws (via media, co-partisans)
- Simple AI chatbots deployed alongside complex chatbots

Key idea: accurate sender **shapes receiver's beliefs about** $\Pr(\theta = a|f_S)$.

The Market for Senders: Design Problem

Formally, type a's design problem is to choose type space $\Theta \subseteq [0, \overline{\gamma}]$ and distribution T.

$$\max_{\Theta, T} p_a \int_{f_S \in F_R} \max_x \mathbb{E}_{f_R | f_S, T, \Theta}[f_R(x)] dP^a_{f_S}(f_S).$$

The key place T enters in is in beliefs over f_R given f_S.

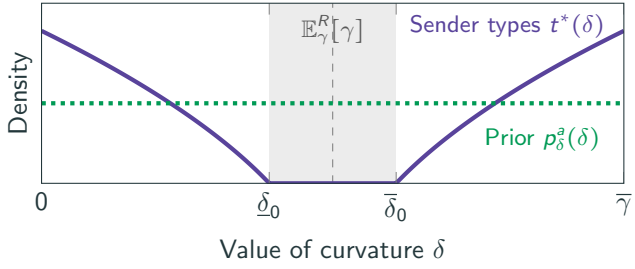
The Market for Senders: Polarization Result

Theorem 3.

There exists a "polarization" function $\pi(\delta)$ such that the distribution of simple types is:

$$t^*(\delta) = p^a_\delta(\delta)\pi(\delta),$$

where p^a_δ is the accurate sender's **prior over curvature values**. *$\pi(\delta)$ is decreasing on an interval $[0, \underline{\delta}_0]$, $= 0$ on $[\underline{\delta}_0, \overline{\delta}_0]$, and increasing on $[\overline{\delta}_0, \overline{\gamma}]$.*



The Market for Senders: Polarization Result, Intuition

Given a model f_S, the accurate sender's gain is given by

$$\frac{(1-p_a)t(\delta)}{(1-p_a)t(\delta)+p_a p_\gamma^R(\delta)}\Big(\mathbb{E}_\gamma^R[\gamma|f_S]-\delta\Big)\Big(\sigma_\delta^2-(x-x_\delta)^2\Big)\geq 0$$

Inequality strict for $\delta \neq \mathbb{E}_\gamma^R[\gamma|f_S]$. Gain is increasing in $|\mathbb{E}_\gamma^R[\gamma|f_S]-\delta|$.

- Largest gain from simple models is at very **high or low** δ values.
- Thus, place more senders near those extreme values.
- In fact, t places 0 mass on an **interval** of intermediate δ values.

Full proof utilizes calculus of variations, more involved (eneral type space, continuity, etc.)

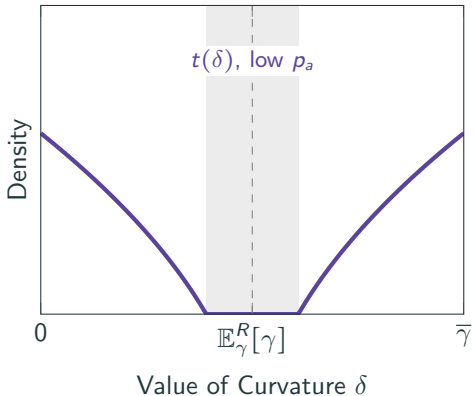
Limiting Cases of Polarization

Recall: p_a is ex-ante prob. of seeing accurate sender. Could be exogenous or endogenous.

Proposition 2.

The polarization of $t^*(\delta)$ is increasing in p_a. As $p_a \to 1$, $t^*(\delta)$ converges to:

- *either a distribution that places mass only on $\delta = 0$ (linear regressions);*
- *a distribution that places mass only on $\delta = \overline{\gamma}$ (worst-case scenarios);*
- *or a distribution that places mass only on $\delta = 0$ and $\delta = \overline{\gamma}$.*



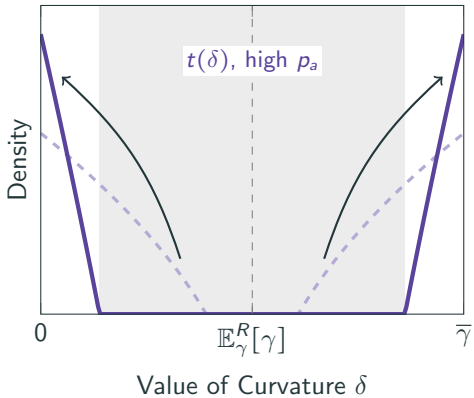
Limiting Cases of Polarization

Recall: p_a is ex-ante prob. of seeing accurate sender. Could be exogenous or endogenous.

Proposition 2.

The polarization of $t^*(\delta)$ is increasing in p_a. As $p_a \to 1$, $t^*(\delta)$ converges to:

- *either a distribution that places mass only on $\delta = 0$ (linear regressions);*
- *a distribution that places mass only on $\delta = \overline{\gamma}$ (worst-case scenarios);*
- *or a distribution that places mass only on $\delta = 0$ and $\delta = \overline{\gamma}$.*



Conclusion

Receiver hires sender to help her make a choie. His information comes in the form of a **model**.

- Model may be **simple**: **fixes curvature** of functional forms at pre-calibrated values.
- Sender wants to make receiver **optimistic** about perceived gains from making choice.

Datasets underlying simple models **poorly suited** to eventual choice.

- Information about an excessively wide range of choices.
- Information about a single, extreme choice .
- Datasets can be extremely collinear.

Accurate model sender gain from being perceived as simple senders.

- We will see both accurate and simple models in the real world.
- Simple models **polarized** into excessively low curvature, excessively high curvature models.

Bonus Material

What else is in the paper?

- Forecast error: R cares not just about the value of choices but their **accuracy**. Second-best form of sampling is σ intermediate (vs. σ low or high).
- Generalized functional forms: attempts to extend beyond quadratic costs.
- General type space: sender type is **interval** of δ values.

References i

Alonso, Ricardo and Odilon Câmara (2016). **"Bayesian persuasion with heterogeneous priors".** In: *Journal of Economic Theory* 165, pp. 672–706.

Esponda, Ignacio and Demian Pouzo (2016). **"Berk–Nash equilibrium: A framework for modeling agents with misspecified models".** In: *Econometrica* 84.3, pp. 1093–1130.

Spiegler, Ran (2016). **"Bayesian networks and boundedly rational expectations".** In: *The Quarterly Journal of Economics* 131.3, pp. 1243–1290.

Heidhues, Paul, Botond Kőszegi, and Philipp Strack (2018). **"Unrealistic expectations and misguided learning".** In: *Econometrica* 86.4, pp. 1159–1214.

Li, Ye and Chen Wang (2018). **"Delegation Uncertainty".** In: *Columbia Business School Research Paper* 15-54.

Deimen, Inga and Dezso Szalay (2019). **"Delegated expertise, authority, and communication".** In: *American Economic Review* 109.4, pp. 1349–1374.

References ii

Henry, Emeric and Marco Ottaviani (2019). **"Research and the approval process: The organization of persuasion".** In: *American Economic Review* 109.3, pp. 911–955.

Eliaz, Kfir and Ran Spiegler (2020). **"A model of competing narratives".** In: *American Economic Review* 110.12, pp. 3786–3816.

Kishishita, Daiki (2020). **"(Not) delegating decisions to experts: The effect of uncertainty".** In: *Journal of Economic Theory* 190, p. 105117.

Mailath, George J and Larry Samuelson (2020). **"Learning under diverse world views: Model-based inference".** In: *American Economic Review* 110.5, pp. 1464–1501.

Bohren, J Aislinn and Daniel N Hauser (2021). **"Learning with heterogeneous misspecified models: Characterization and robustness".** In: *Econometrica* 89.6, pp. 3025–3077.

Di Tillio, Alfredo, Marco Ottaviani, and Peter Norman Sørensen (2021). **"Strategic sample selection".** In: *Econometrica* 89.2, pp. 911–953.

Fudenberg, Drew, Giacomo Lanzani, and Philipp Strack (2021). **"Limit points of endogenous misspecified learning".** In: *Econometrica* 89.3, pp. 1065–1098.

References iii

Schwartzstein, Joshua and Adi Sunderam (2021). **"Using models to persuade".** In: *American Economic Review* 111.1, pp. 276–323.

Cheng, Haw and Alice Hsiaw (2022). **"Distrust in experts and the origins of disagreement".** In: *Journal of economic theory* 200, p. 105401.

De Clippel, Geoffroy and Xu Zhang (2022). **"Non-bayesian persuasion".** In: *Journal of Political Economy* 130.10, pp. 2594–2642.

Gleyze, Simon and Agathe Pernoud (2022). **"The Value of Model Misspecification in Communication".** In.

Levy, Gilat, Inés Moreno de Barreda, and Ronny Razin (2022a). **"Persuasion with correlation neglect: a full manipulation result".** In: *American Economic Review: Insights* 4.1, pp. 123–138.

Levy, Gilat, Ronny Razin, and Alwyn Young (2022b). **"Misspecified politics and the recurrence of populism".** In: *American Economic Review* 112.3, pp. 928–962.

References iv

Montiel Olea, José Luis et al. (2022). **"Competing models".** In: *The Quarterly Journal of Economics* 137.4, pp. 2419–2457.

Aina, Chiara (2023). **"Tailored stories".** In: *Unpublished manuscript* 12, p. 52.

Barron, Kai and Tilman Fries (2024). **Narrative persuasion.** Tech. rep. WZB Discussion Paper.

Aina, Chiara and Florian Schneider (2025). **"Weighting competing models".** In.

Bohren, J Aislinn and Daniel N Hauser (2025). **"Misspecified models in learning and games".** In: *Annual Review of Economics* 17.1, pp. 427–451.

Eliaz, Kfir, Simone Galperti, and Ran Spiegler (2025). **"False narratives and political mobilization".** In: *Journal of the European Economic Association* 23.3, pp. 983–1027.

Kolotilin, Anton and Andriy Zapechelnyuk (2025). **"Persuasion meets delegation".** In: *Econometrica* 93.1, pp. 195–228.

Dai, Yifan, Drew Fudenberg, and Harry Pei (2026). **"Bayesian Persuasion with Selective Disclosure".** In: *arXiv preprint arXiv:2601.05914*.